PalmSource™ 99

Developing C/C++ Conduits for Macintosh

Geoff Richmond
Sr. Development Support Engineer

3Com
More connected.

Overview

PalmSource 99

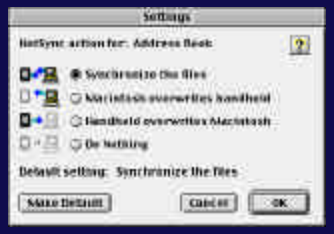
- Synchronization Overview
- HotSync® Architecture
- Choosing the Right Development Tool
- Palm OS® Conduit Project Stationery
- Conduit Development for Macintosh
- Installing Conduits on Mac OS
- Summary

Synchronization Overview

PalmSource 99

- Desktop Synchronization
  - Handheld and synchronization engine on local workstation
  - Data usually located on local workstation
- Network Synchronization
  - Handheld connected to any computer on network
  - Synchronization runs on handheld user's workstation
  - Data accessible from handheld user's workstation
- Server Synchronization
  - Handheld connected directly to network (or client)
  - Synchronization runs on servers
  - Data and synchronization engines distributed

Synchronization Dataflow







Mirror image	Ideal for applications that run on both platforms and allow modifications on both; makes data identical on both platforms. (address book and date book applications)	Your conduit must provide conflict resolution solutions when the same record has been modified on both platforms. This must occur without user intervention.
One directional Upload / Download	For applications that run on both platforms but only allow data modification on one; copies the data to the other platform.	Requires less time, and should be used when possible
Transaction based	For applications that need to perform additional processing on the desktop computer between record synchronizations. (e-mail, web, PQA, etc.)	Only use this when required, because it slows down the HotSync® process.
Back up	For applications that don't have desktop components; simply copies the handheld data to the desktop.	Can use the Backup conduit provided by Palm Computing.

HotSync® Architecture

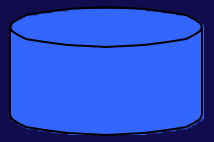
HotSync Manager



Application-Specific Conduit



Data Source



Conduit API
- OpenConduit()
- ConfigureConduit()

Conduit Base Classes
- CSynchronizer
- CRecordIterator
- CSyncRecord

- Local/networked data file
- Local/networked database
- Web-based data source
- E-mail or news server
- Dial-in connection

SyncManager API
- SyncOpenDb()
- SyncReadNextModified()
- SyncWriteRec()

Handheld



Contains applications/databases
- Unique creator id for each application
- Built-in HotSync device client
- HotSync process initiated by the device

Conduit Development Environments

PalmSource 99

- CDK 2.2 for Macintosh
  Palm Computing ®
- CDK 3.0 for Windows
  Palm Computing
- CDK/J for Java
  Palm Computing



http://www.palm.com/devzone/conduits.html

- Satellite Forms ®
  Puma Technology ®
- Scout Architect ®
  Riverbed Technologies ®





Features of Palm OS® Conduit Development Kit for Macintosh

PalmSource 99

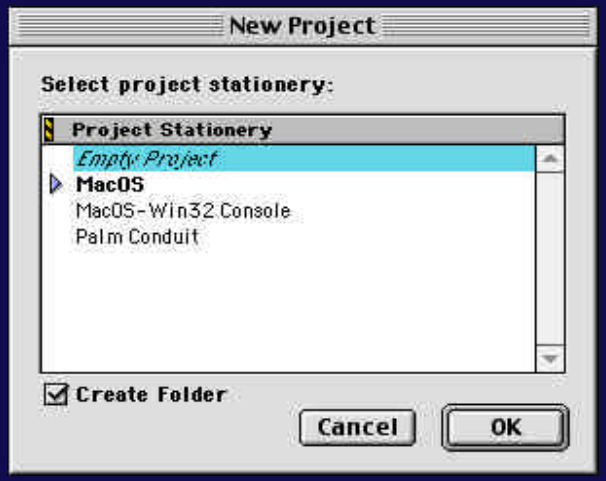
- Palm Conduit Project Stationery
  - Project Starter Code Generation
- Conduit Base Classes
  - core synchronization logic
- Conduit Sample Projects
  - source code for Palm standard applications

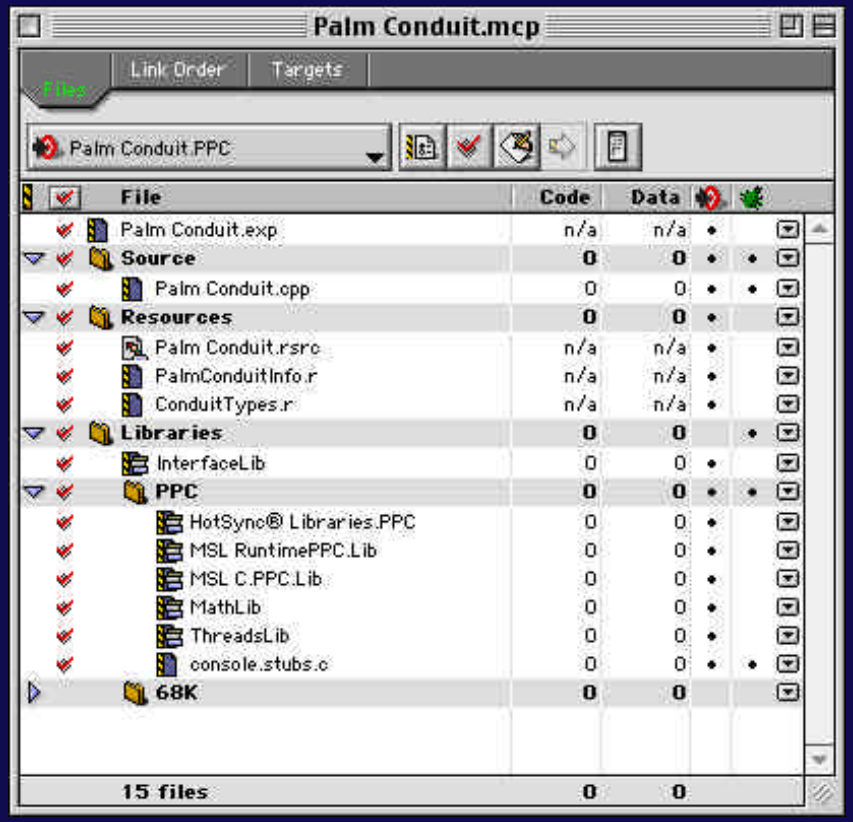
Features of Palm OS® Conduit Development Kits

- Install Aide API
  - Permits easy installation/management of Palm Computing Platform applications
- SyncManager API
  - Database/record access

Palm OS® Conduit Project Stationery

PalmSource 99





Easy configuration of conduit starter code

- Metrowerks CodeWarrior ® 3 or 4
- Generates starter code for Mac® OS conduits
- Generic Conduit to be supported in future releases

Conduit Entry Points

PalmSource 99

Required

- OpenConduit()
  - Synchronization entry point
- GetConduitName()
  - Returns name of conduit for display purposes
- GetConduitVersion()
  - Returns conduit version
- GetActionString()
  - Returns the string seen by the user in ‘Next HotSync action’ column of HotSync® Manager actions dialog
- ConfigureConduit()
  - Presents a dialog that allows the user to configure the conduit

SyncManager API

PalmSource 99

Database/Record Structure

Database Name
Database Creator
Database Type
Number of Records
Last Sync Date
AppInfo Block Pointer
Sort Info Pointer
Data Record 1
Data Record 2
Data Record n

Use SyncManager API to access remote databases and records.

SyncReadNextModifiedRec()

CRawRecordInfo Class

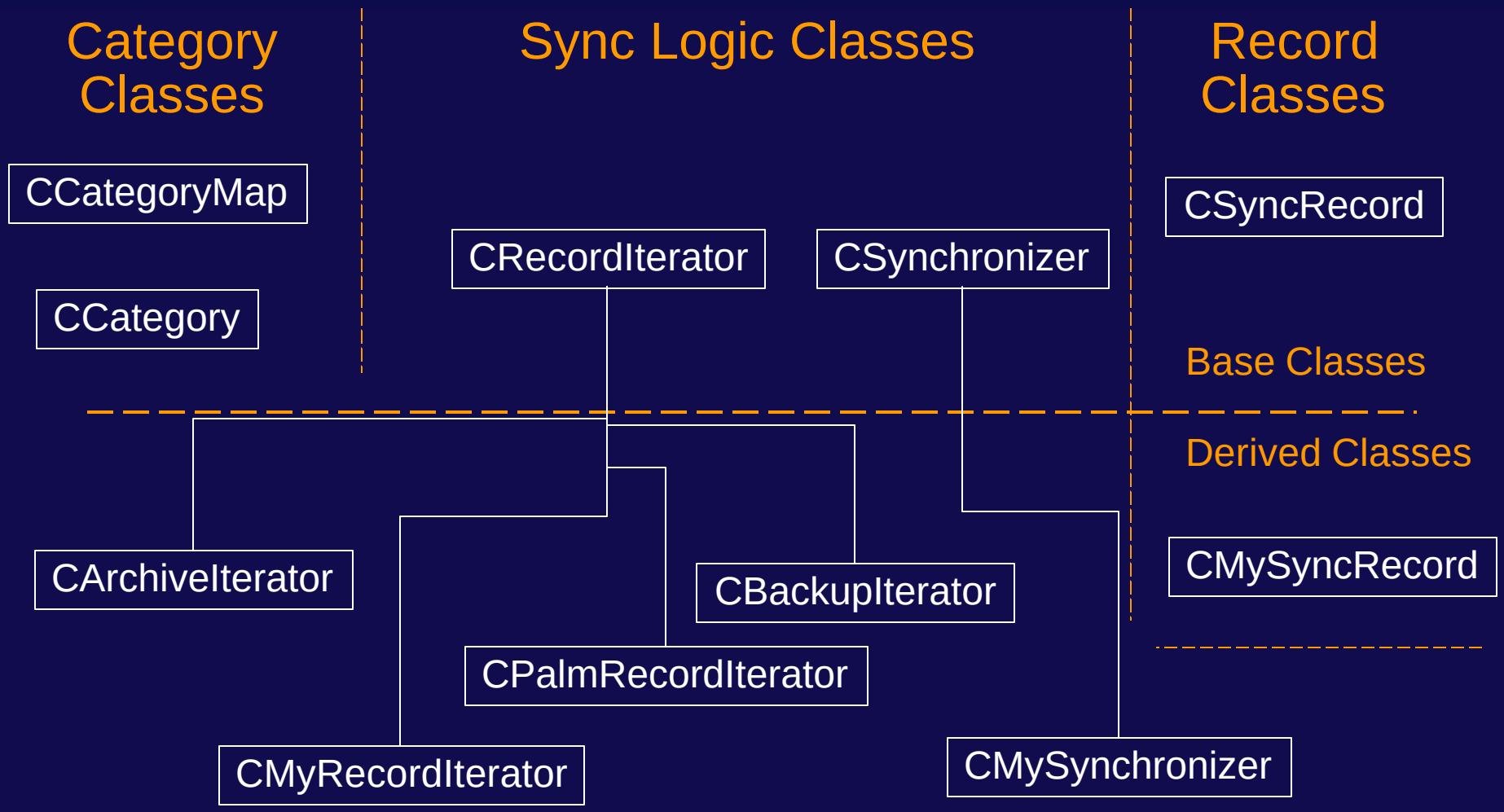
m_RecId
m_RecIndex
m_attribs
m_CatId
m_pBytes



Unique Id	Category	Attribs	Raw Record Data

Palm OS® Conduit Class Hierarchy

PalmSource 99



Native Synchronization Objects

- CSynchronizer
  - CSynchronizer is the main synchronization engine
- CRecordIterator
  - CRecordIterator is an abstract data manipulation class that you implement to iterate through a database. You must implement this class to create a conduit
  - You override methods in this class to provide access to your data, returning the data in CSyncRecord objects
- CSyncRecord
  - CSyncRecord is a record format class

Conduit Control Flow

PalmSource 99

- HotSync® Manager calls your OpenConduit() function
  - Examine SyncProperties structure for sync information
- Your OpenConduit() function instantiates a CSynchronizer object and performs the following tasks:
  - Register the conduit
  - Create the Iterator objects (handheld, desktop, archive)
  - Synchronize categories information using Native Sync logic
  - Synchronize application information (AppInfo block)
  - Synchronize records using Fast Sync, Slow Sync, HH overwrites desktop, Desktop overwrites HH, or No Action
  - Delete the Iterator objects
  - Log results to HotSync log

Conduit Development Steps

PalmSource 99

Creating a Conduit from Palm OS® Conduit Classes

1. Define your record structure in CMyRecordIterator.h.

   The desktop iterator supports records as structures. All data elements should be declared public for easy access.

```
struct MemoItem {
        UInt32 internalRecID;
        UInt32 recordID;
        UInt8 categoryID;
        UInt32 attributes;
        char   memo[MEMO_TEXT_SIZE];
};

typedef struct MemoItem MemoItem;
typedef TArray<MemoItem>                CMemoItemArray;
typedef TArrayIterator<MemoItem>        CMemoItemArrayIterator;
```

2. Convert to/from your record structure in CMyRecordIterator and CPalmRecordIterator.

```
virtual void ConvertFromSyncRecord (const CSyncRecord& inCSyncRecord, void* outRecordData);
virtual void ConvertToSyncRecord (const void* inRecordData, CSyncRecord* outCSyncRecord);
```

Conduit Development Steps (Cont.)

Creating a Conduit from Palm OS® Conduit Classes

3. Read and write your data in PreSync() and PostSync() methods, respectively.

The desktop iterator reads and writes the records on the desktop.

```
void CMemoTestRecordIterator::PreSync()
{
      this->ReadDataFile();
}

void CMemoTestRecordIterator::PostSync()
{
     this->WriteDataFile();
}
```

The ReadDataFile() and WriteDataFile() methods are easily customized for any data format (CSV, ODBC, others). The samples provided with the CDK store records in an ASCII text file format. Data is stored as an array of MemoItems in the CMemoTestRecordIterator class.

The category data is also stored in the data file.

Data is read/written in a single operation.

Conduit Development Steps (Cont.)

Creating a Conduit from Palm OS® Conduit Classes

4. Create CBackupIterator and CArchiveIterator(s) as sub-classes of CRecordIterator.

   Performs the Storing and Retrieving of backup and archive files.
   Customize the filenames.

5. Create CMySynchronizer as a sub-class of CSynchronizer and implement the MakeRemoteIterator() and MakeLocalIterator() methods.

6. Optionally override the SynchronizeAppInfoBlock() and SynchronizeSortInfoBlock() methods for your application.

Conduit Debugging Hints

PalmSource 99

- If your conduit does not run or does not appear in ‘Custom Dialog’ box, refer to “Palm Programmer’s Companion for Macintosh” for troubleshooting help
- Visit the Knowledge Base for debugging procedures. (http://www.palm.com/devzone)
- Disable automatic power-off via ‘Shortcut.3’
- Disable Serial Link timeouts while single-step debugging. (See “Palm Programmer’s Companion for Macintosh” for additional information)

Installing Conduits

PalmSource 99

- Use Palm-supplied API's for application and conduit installation
  - Install Aide API (PltInstallFile())
- Visit Knowledge Base for sample install code

Summary

PalmSource 99

- The Palm Computing® Platform is the leader in handheld computing and information management
- Synchronization (Data Exchange) plays an important role in handheld computing. (personal and enterprise information management)
- Successful implementations depend on choosing the right synchronization strategy
- The HotSync® architecture is adaptable to many synchronization scenarios
- The Palm Conduit Project Stationery provides a flexible framework for development of C++ Conduits for Macintosh
- The Palm Computing® CDK provides a complete set of tools for conduit development for Macintosh


PalmSource™ 99
3Com
More connected.